Clinical Gastroenterology and Hepatology 2026;24:296–305

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Evaluation and Management of Refractory Constipation: Expert Review

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DESCRIPTION: **Although most patients with chronic constipation respond to medical therapy, a subset experiences refractory constipation (RC), which poses unique diagnostic and therapeutic challenges. Because RC is relatively rare, clinicians should systematically (1) exclude correctable secondary causes such as medications, neurologic disorders, and defecatory disorders (DDs); (2) confirm the presence of slow colonic transit; and (3) ensure that patients have undergone adequate trials of over-the-counter and Food and Drug Administration–approved medications and nonpharmacologic therapies, including combinations thereof. Surgical treatments, such as colectomy, may be considered in patients who fail available treatments. However, surgical treatment of chronic constipation is associated with increased risk of complications and a not insignificant number of unsatisfactory outcomes. Prior to advising surgery, it is essential to confirm slow colonic transit, exclude concurrent DDs, and evaluate for severe, symptomatic delays in gastric emptying or small bowel dysmotility. Psychological comorbidities may exacerbate symptoms and adversely affect surgical outcomes. Hence, preoperative psychological evaluation is also advisable to assess suitability for surgery. Relative contraindications to surgical treatment of RC include clinically significant upper-gut dysmotility, severe, untreated psychiatric disease, and predominant complaints of bloating and/or abdominal pain. In uncertain cases, a temporary diverting loop ileostomy may help predict the potential response to colectomy. A colectomy with ileorectal anastomosis should only be offered to patients without ongoing DDs.**

METHODS: **This expert review was commissioned and approved by the American Gastroenterological Association (AGA) Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership, and underwent internal peer review by the Clinical Practice Updates Committee and external peer review through standard procedures of Clinical Gastroenterology and Hepatology. These Best Practice Advice statements were drawn from a review of the published literature and from expert opinion. Because systematic reviews were not performed, these Best Practice Advice statements do not carry formal ratings of the quality of evidence or strength of the presented considerations.**

Chronic constipation (CC) affects 8%–12% of the US population,[1] with 3 million patients/year seeking clinical evaluation.[2] While the terms *chronic constipation*, *chronic idiopathic constipation*, and *functional constipation* are interchangeable, we use CC throughout this review. Though most patients with CC respond to standard treatments, a subset remains refractory to available therapies and merit the designation of "refractory." Such patients consume significant healthcare resources and frequently present to tertiary care centers. Colectomy as treatment of refractory constipation (RC) is associated with high perioperative healthcare resource utilization with increased risk of significant complications including small bowel obstruction, recurrent constipation, residual abdominal pain and bloating, and ongoing need for laxatives.[3,4]

This review is designed to provide best practice advice on the diagnosis, treatment, and medical and surgical management of patients with RC. Even with careful assessment as outlined subsequently, the decision to pursue surgery for RC should be made by an experienced clinician based on the preponderance of available evidence for a given patient without adhering

Abbreviations used in this paper: ARM, anorectal manometry; BET, balloon expulsion testing; CC, chronic constipation; DD, defecatory disorder; MR, magnetic resonance; RC, refractory constipation; SNS, sacral nerve stimulation.

Most current article

1542-3565/$36.00
https://doi.org/10.1016/j.cgh.2025.09.031

to a strict algorithm. We have developed Best Practice Advice statements to address 14 key clinical issues.

Best Practice Advice 1. RC is defined by infrequent and/or unsatisfactory bowel habits, with or without abdominal pain, despite an adequate trial of lifestyle, dietary, medical, and/or pelvic floor biofeedback therapy when indicated in adults who satisfy criteria for CC or constipation-predominant irritable bowel disease.

Defining RC is challenging due to the interplay of multiple factors influencing bowel habits including colonic and anorectal function, as well as partly reversible factors such as diet, physical activity, medications, and comorbid physical and psychological conditions. Other complexities in defining RC arise from variability of treatment(s) considered necessary before making the diagnosis and differences in how patients describe and perceive constipation. International consensus guidelines define RC as "infrequent and/or unsatisfactory bowel habits" and any of the following[5]:

1) Inadequate number of bowel movements most of the time and complete bowel movements occur on fewer than 3 days/week.
2) Straining on most occasions or worsening straining.
3) No improvement in stool consistency on current therapy and Bristol stool scale score <3.
4) Insufficient improvement of another sign or symptom of CC on current treatment.
5) Poor tolerability of the current treatment makes the relief provided unacceptable.

A practical definition of RC is essential for developing a tailored, pathophysiology-driven approach to manage the symptom while avoiding surgery in which it is not indicated and potentially harmful. Our proposed definition and management strategies are guided by the underlying pathophysiology, using objective measures where possible.

Best Practice Advice 2. Most patients with CC should be evaluated with anorectal manometry (ARM) with balloon expulsion testing (BET) and complete a course of pelvic floor biofeedback therapy, when indicated, before being labeled as having RC.

Approximately 37% of CC patients undergoing pelvic floor testing have dyssynergic defecation (DD),[6] which is defined by constipation and impaired rectal evacuation due to inadequate rectal propulsive forces and/or inadequate anal relaxation during defecation.[7] DD should be diagnosed and treated before diagnosing RC. Prior to physiologic testing in CC, a careful rectal examination should be performed in every patient to evaluate likelihood of DD and other anorectal pathology.[8] An ARM with BET is generally sufficient to diagnose DD and can also predict which patients are likely to benefit from pelvic floor biofeedback therapy,[9] which is the cornerstone of managing DD.[10] DD can be diagnosed based on a reduced rectoanal gradient on ARM, prolonged BET ($\geq$1 minute), or inadequate rectal evacuation during defecography.[10,11] Other features of DD include an elevated anal resting pressure and reduced anal relaxation.[12] Two out of 3 abnormal tests are typically required to make a diagnosis of DD.

Biofeedback therapy can improve rectal evacuation and coordination between the abdominal wall and anorectum in patients with DD.[10,13] While the duration of biofeedback therapy varies, it often requires up to 6 months of consistent practice for benefits to manifest. Normalization of BET after biofeedback therapy suggests the resolution of DD.[9]

Best Practice Advice 3. Patients with RC should be thoroughly evaluated for secondary causes of constipation such as medications, disordered eating, or comorbid neurological diseases.

Medications are among the most common iatrogenic causes of CC including opioid-induced or opioid-exacerbated constipation. Other frequent secondary causes include anticholinergic agents such as antipsychotics and iron supplements. In many patients, these agents exacerbate preexisting constipation rather than being the primary cause.[14] Inadequate caloric intake may also predispose to RC, particularly in individuals with eating disorders such as anorexia nervosa, which are disproportionately represented in gastrointestinal clinics. Some patients with RC may have an undiagnosed or unrecognized eating disorder driving their symptoms.[15,16]

Occasionally, CC may coexist with conditions such as endometriosis, or rarely, undiagnosed or early-stage neurological diseases like Parkinson's disease and multiple sclerosis, without the typical neurologic manifestations.[17,18] Evaluating autonomic nervous function may be considered in CC patients presenting with orthostatic symptoms.[19] Alternative therapeutic options are available for patients with RC who also have autonomic dysfunction (see Best Practice Advice 8).

Best Practice Advice 4. Clinicians caring for RC patients should describe its typical course and set realistic expectations.

Many patients with CC have misconceptions about normal bowel habits. In the United States, approximately 95% of individuals have between 3 bowel movements/day and 3 bowel movements/week, with 90% having an average stool consistency on the Bristol stool scale between 3 and 5 for men and 2 and 6 for women.[20] Additionally, many patients fail to recognize that CC is a chronic disease requiring ongoing treatment. They should be counselled to expect likely lifelong medical treatment, with treatment goals guided by normative bowel function data.[20] Educating patients about these goals will help align expectations with outcomes and reinforce the importance of adherence to ongoing therapy.

Best Practice Advice 5. Patients with RC should undergo colonic transit testing off of treatments to document slow-transit constipation and ideally also on a maximal laxative regimen to document the refractory nature of symptoms.

Currently, colonic transit may be assessed with scintigraphy or radiopaque markers. There are several options using radiopaque markers with varying degrees of simplicity, radiation exposure, and ability to quantify colonic transit time.[21] These methods have been extensively validated, entail minimal radiation exposure, and can quantify overall and regional colon transit. Two wireless motility capsules are also now available to measure regional gut transit (see Best Practice Advice 11).[22]

Colonic transit should be evaluated after patients have stopped laxatives for at least 7 days and medications that delay colon transit for at least 2 weeks.[7,23,24] However, given the poor intraindividual day-to-day reproducibility in patients with DD or colonic inertia,[25] repeat testing may be necessary, and in some patients, while taking laxatives. There is a significant overlap between slow transit constipation and DD that may be as high as 40%; mechanistically, stool in the rectosigmoid colon may activate rectocolonic inhibitory reflexes delaying colon transit. Normal colon transit on laxatives suggests that symptoms are due to other factors such as DD, visceral hypersensitivity, or laxative side effects.[7] In contrast, slow transit suggests that colonic dysmotility is a contributing factor, and colectomy may be considered in select cases if a defecatory disorder has been excluded, as discussed subsequently.

Best Practice Advice 6. Among patients with RC, defecography should be considered.

Barium or magnetic resonance (MR) defecography, providing real-time imaging of anorectal and pelvic floor structures during defecation,[26,27] should be considered when clinical features suggest a structural abnormality or when ARM and BET are inconclusive.[7,23] Abnormalities that can be identified on defecography include inadequate or excessive widening of the anorectal angle and/or perineal descent during defecation.[27,28] Internal intussusception, solitary rectal ulcers, rectoceles, and rectal prolapse as well as enteroceles, bladder, and uterovaginal prolapse may also be identified.[26] Each abnormal anorectal test finding (ie, lower rectoanal gradient by manometry, a prolonged BET, and reduced evacuation by defecography) is highly specific for diagnosing DD.[11] However, in select patients, defecography may help confirm the diagnosis, particularly in patients with an abnormal gradient on ARM and a normal BET.

The choice of barium or MR defecography depends on several factors. When defecography is primarily indicated to assess rectal evacuation, barium defecography—performed in the seated position—may be preferable to MR defecography, which is performed in the supine position. By contrast, MR defecography avoids radiation exposure, offers more precise assessments of pelvic organ prolapse and pelvic floor motion,[29] and is especially useful for identifying pelvic floor dysfunction in patients who have clinical features of DD and a normal BET; this group includes more than 90% of patients with a large rectocele, enterocele, and/or peritoneocele.[27] These anatomic findings are more likely be clinically relevant in selected patients (eg, patients with a normal ARM and impaired contrast evacuation on defecography or patients with a rectocele >4 cm in size). However, MR defecography is less widely available and more expensive.

Best Practice Advice 7. Patients with RC should trial all standard over-the-counter and Food and Drug Administration–approved therapies for constipation as standalone therapy or in combination when accessible.

Studies have shown prucalopride and linaclotide are effective in CC patients unresponsive to over-the-counter laxatives,[30,31] while plecanatide has shown efficacy in "severe" constipation.[32] However, these trials lack rigorous evaluation of prior over-the-counter therapies or colonic transit.

Empiric use of combination therapy, involving agents with different mechanisms of action, is common when monotherapy fails (Figure 1), although data on combination therapies remain anecdotal, for example, combining an osmotic laxative (eg, polyethylene glycol) with a stimulant (eg, bisacodyl) or prokinetic (eg, prucalopride). Bisacodyl stimulates high-amplitude propagating contractions and appears to be particularly effective for increasing spontaneous bowel movements.[33,34] Selected patients may benefit from regular stimulant laxative use. Concerns about the chronic use of stimulant laxatives in managing refractory adult constipation are unfounded.[35] Although Food and Drug Administration guidelines recommend a maximum senna dose of 35 mg twice daily, even up to 1.0 g/day for up to 8 weeks has been safe.[36] For comparison, widely available products like Smooth Move tea often exceed this dose without reported adverse events. Suppositories and enemas can be beneficial, particularly in pelvic floor dysfunction, though there are limited data guiding their use.

Bowel cleanouts may be helpful in clarifying the contribution of constipation to symptoms or prior to starting maintenance treatments—particularly in patients with a large stool burden. However, frequent bowel cleanouts should be used sparingly to avoid the risk of setting unrealistic patient expectations for repeat treatments.

Best Practice Advice 8. We suggest trials of off-label prescription agents (eg, pyridostigmine) for patients with RC who have failed available over-the-counter and Food and Drug Administration–approved agents.

Pyridostigmine, a cholinesterase inhibitor, improves symptoms and accelerates colonic transit in autonomic neuropathy,[37] diabetes mellitus-related constipation,[38] and postural orthostatic tachycardia syndrome.[39] Additionally, preliminary evidence suggests that combining pyridostigmine with bisacodyl may improve outcomes in RC.[38,40] Tenapanor, a sodium hydrogen exchanger 3

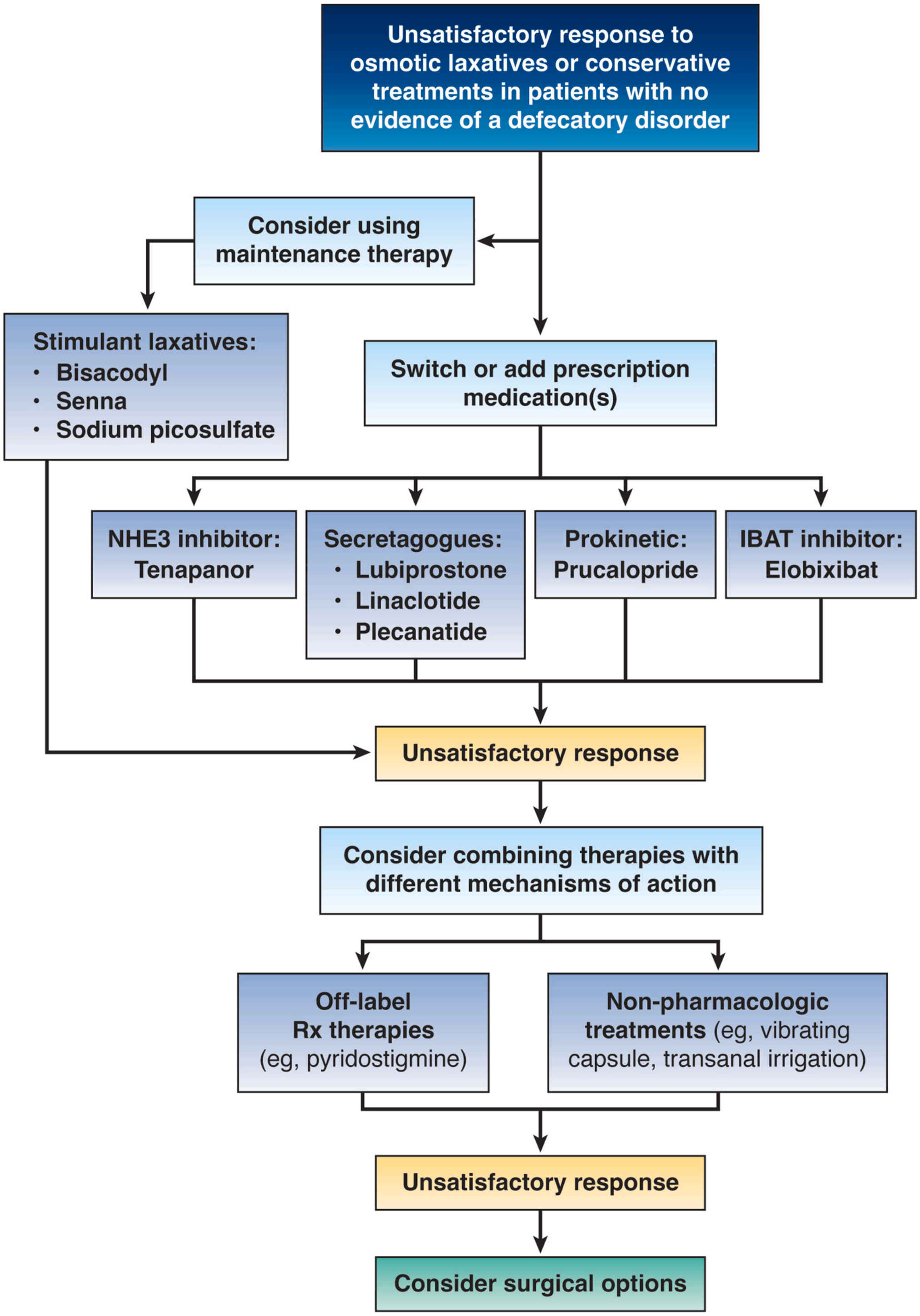

Figure 1. Suggested algorithm for the medical treatment of RC. IBAT, ileal bile acid transporter; NHE3, sodium/hydrogen exchanger 3; Rx, prescription.

inhibitor approved for the treatment of irritable bowel disease with constipation, may also be considered when on-label agents have failed. The ileal bile acid transporter inhibitor elobixibat—approved in several Asian countries—may also be considered where available. Colchicine and misoprostol may improve constipation symptoms. However, given their limited efficacy and unfavorable side effect profile, we do not advise their routine use.

Best Practice Advice 9. Trials of adjunct, nonpharmacologic approaches including transanal irrigation for RC with neurogenic bowel dysfunction, a vibrating capsule, or electroacupuncture are reasonable in patients with RC.

Nonpharmacologic treatments can provide meaningful relief for patients with RC, and may reduce the need for invasive procedures such as colectomy.

Transanal irrigation can be effective, particularly in patients with neurogenic bowel dysfunction,[41] as it reduces stool burden, improves bowel emptying, and enhances quality of life. While sacral nerve stimulation improves symptoms in some patients with fecal incontinence, it did not improve symptoms in CC.[42]

Among patients with CC, including those with "severe" symptoms as defined by no complete spontaneous bowel movements over a 2-week baseline period, a vibrating capsule has been shown to improve symptoms compared with placebo.[43–45] Electroacupuncture, which combines traditional acupuncture with electrical stimulation, has been demonstrated to be more effective than sham acupuncture and comparable to prucalopride.[46,47]

Integrating these nonpharmacologic approaches with tailored pharmacologic treatments offers the potential for a holistic, individualized management strategy for RC. However, further studies are needed to evaluate the effectiveness of combined pharmaco- and non-pharmacological therapies.

Best Practice Advice 10. Surgical therapy for RC should only be considered after confirming slow-transit constipation and excluding pelvic floor dysfunction. Only patients with RC and no evidence of an ongoing DD should be offered a colectomy with ileorectal anastomosis.

Proper patient selection is crucial for achieving favorable surgical outcomes for RC (Figure 2). Outcomes of colectomy with ileorectal anastomosis are better in patients who have slow colonic transit without DD.[48] However, there is a significant overlap between slow transit constipation and DD, that may be as high as 40%[6,49–51]; mechanistically, stool in the rectosigmoid colon may activate rectocolonic inhibitory reflexes delaying colon transit.[52,53] While the studies examining the impact of DD on colectomy outcomes are inconsistent,[54–57] these inconsistencies are largely due varying methodologies for diagnosing DD, differences in interventions for DD (biofeedback before or after surgery), and the reliance on small retrospective reviews. Regardless, there is ample evidence supporting the need to address DD prior to colectomy and ileorectal anastomosis.[48,58] Biofeedback therapy should be trialed prior to colectomy, as it may normalize colonic transit in many patients with DD.[59] Failure to address DD prior to colectomy with ileorectal anastomosis can lead to poor functional outcomes such as persistent abdominal pain and constipation,[58] contributing to low satisfaction rates.[54] Thus, there is strong consensus on avoiding colectomy with ileorectal anastomosis in patients with DD given this interplay between slow transit constipation and DD.

Best Practice Advice 11. Preoperative evaluation should include regional gut transit testing.

A subset of patients with RC also experience upper gastrointestinal symptoms[60] and/or gastrointestinal dysmotility.[61,62] A meta-analysis found that recurrent constipation and poor quality of life after colectomy were more common in patients with upper gastrointestinal dysmotility before surgery.[61,63] Hence, we suggest that gastric emptying should be obtained to identify moderate-to-severe delayed gastric emptying[64] or more diffuse upper gastrointestinal dysmotility in constipated patients with clinically significant upper gastrointestinal symptoms.[48,65] Constipation itself can contribute to upper gastrointestinal symptoms, which, in our experience, may improve following colectomy. Neither barium x-ray nor lactulose breath tests, which have been inadequately validated and suffer from several limitations, should be used to assess small intestinal transit,[66] but some small bowel imaging (including small bowel follow through or computed tomography/MR enterography) may be helpful to exclude small intestinal dilatation in those for whom there is a high index of suspicion for small intestinal dysmotility or chronic intestinal pseudo-obstruction. Whole-gut scintigraphy, which is well validated but used in only a few centers—or the newly approved wireless motility capsules—provide alternative means of assessing regional gut transit times in this patient population. Prior to colectomy, intraluminal assessments of colonic motility with manometry and/or a barostat, if available, should be considered especially when the colonic transit is equivocal.[67–69] Of note, fasting and/or postprandial colonic tone and/or compliance were reduced in 40% patients with normal and 47% with slow transit constipation.[70] These findings suggest that some patients with slow colon transit have normal motility and vice versa.

Best Practice Advice 12: Patients should undergo a psychological evaluation before surgical therapy for RC.

Psychological comorbidities such as anxiety and depression are common among patients with CC and are particularly common in those with DD.[71–73]Psychological comorbidities often exacerbate symptom severity and burden,[74–76] reduce quality of life,[77] and increase demand for surgical interventions. However, these psychological factors also adversely affect surgical outcomes.[4,48,78] Additionally, psychological complications, including depression, have been reported postoperatively.[4,55] These considerations highlight the importance of a thorough psychological evaluation with a professional knowledgeable about the demands of surgery to guide patient selection prior to surgery to enhance surgical success and minimize the risk of adverse events.

Best Practice Advice 13. Relative contraindications to surgical treatment of RC include severe, untreated psychiatric disease including active eating disorders and unresolved issues related to sexual trauma; primary complaints of bloating and/or abdominal pain; and reversible, secondary causes of constipation.

Recognizing comorbid eating disorders is crucial, as they affect up to 19% of patients with CC[15] and increasing calorie intake can normalize colonic transit.[79] This underscores the importance of preoperative screening. Up to 40% of patients with CC and 20% of those with DD have a history of sexual abuse,[73,80] which is associated with more severe symptoms, likely because abuse is associated with heightened gastrointestinal-

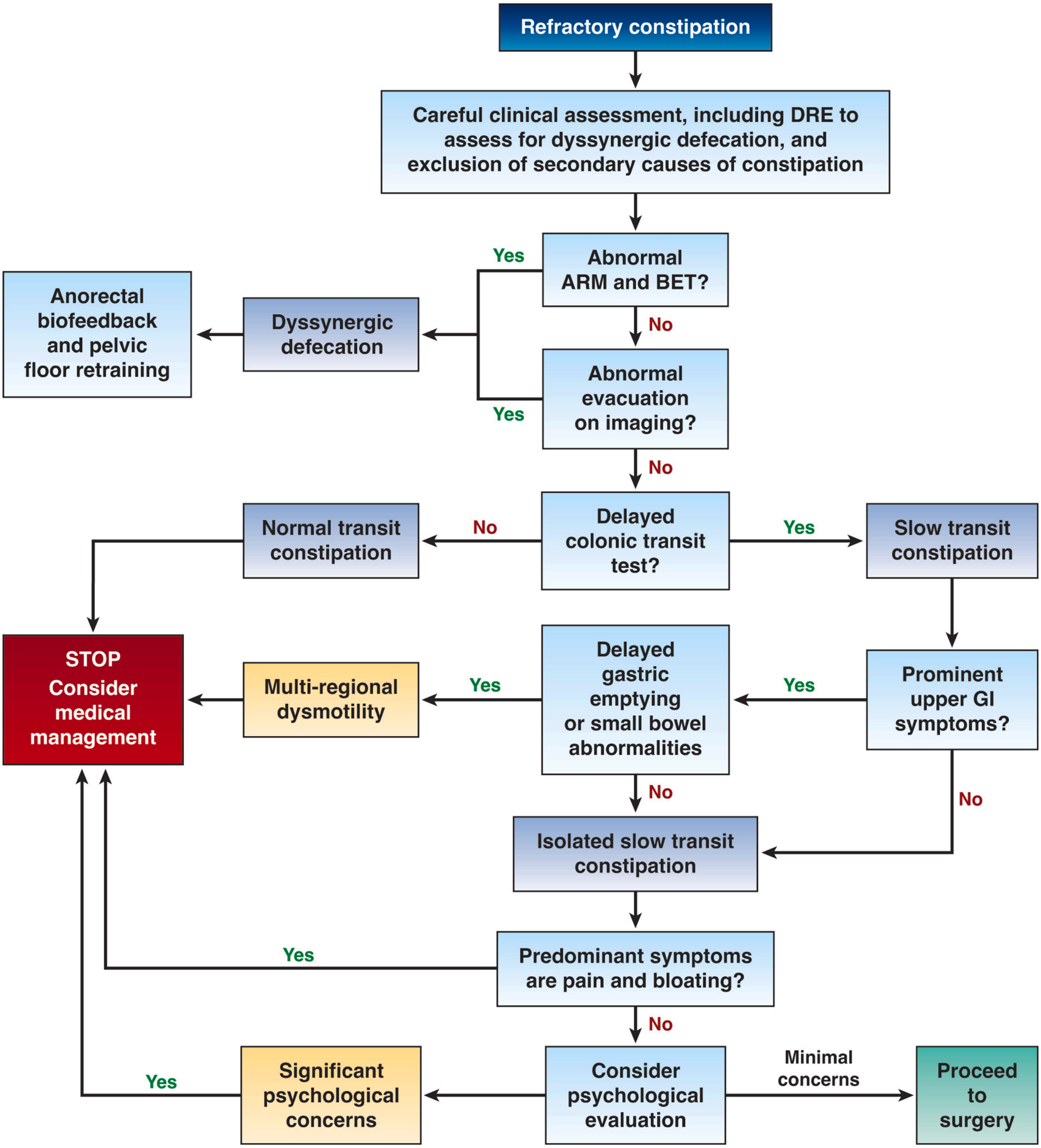

Figure 2. Suggested algorithm for the proper selection of patients for the surgical treatment of RC. DRE, digital rectal examination.

specific anxiety, depression, and rectal sensitivity.[81] Also, a history of abuse strongly predicts adverse outcomes and increased healthcare related to abdominal complaints after colectomy.[82]

Chronic abdominal pain, bloating, and fecal incontinence can persist in up to 50% of patients with CC postcolectomy and are linked to increased healthcare utilization and poor quality of life.[83,84] Therefore, patients with prominent abdominal symptoms should be carefully screened prior to colectomy.[48,85,86]

In addition to the relative contraindications listed previously, we note that outcomes of colectomy with ileorectal anastomosis are improved in patients with proven slow colonic transit.[49] Emerging evidence suggests occurrence of concurrent multiregional gastrointestinal dysmotility in patients with slow transit

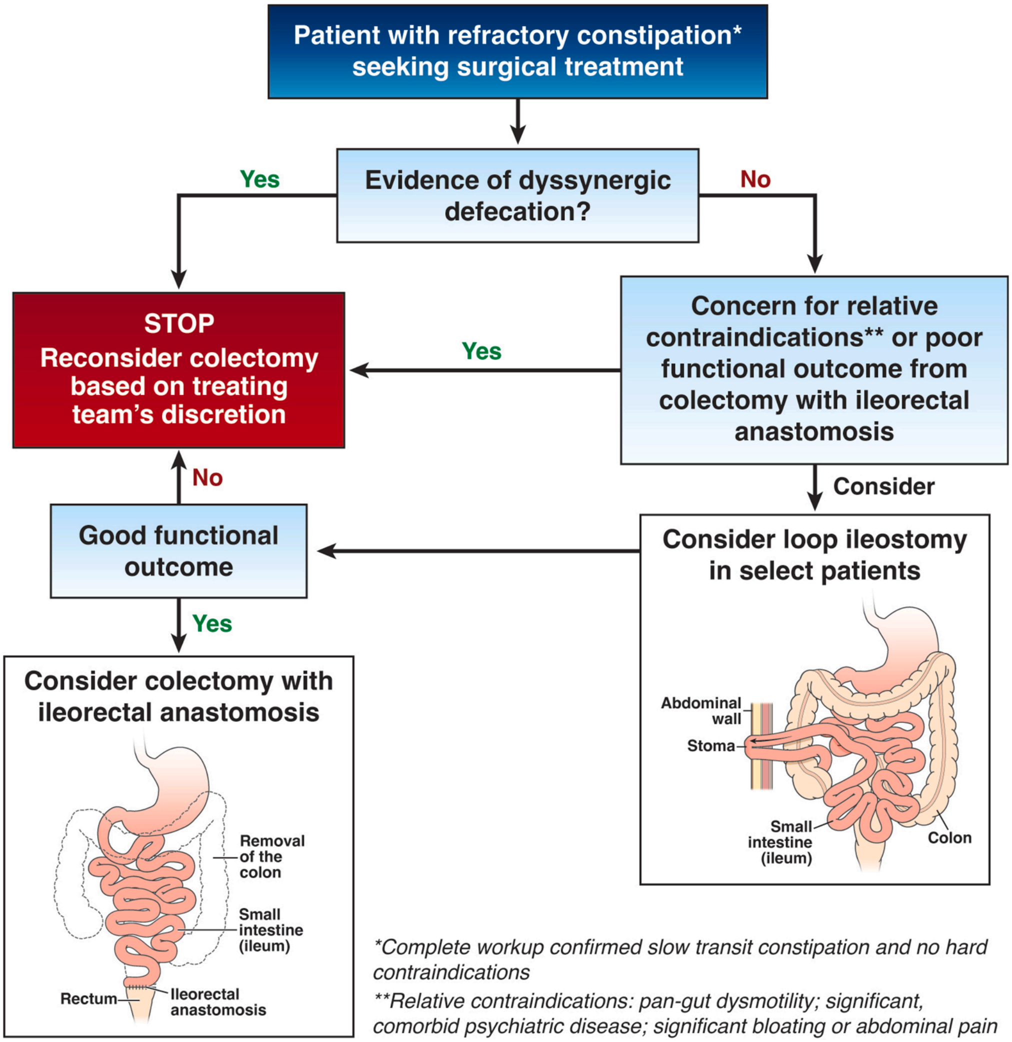

Figure 3. Suggested algorithm for selection of the type of surgical procedure for the treatment of RC.

constipation,[64,87] which correlates with increased surgical complications and poor outcomes and[63,88,89] represents a relative contraindication for surgery.

Best Practice Advice 14. A diverting loop ileostomy offers a diagnostic trial of surgical treatment of RC, especially where there are concerns about efficacy or relative contraindications.

Colectomy with ileorectal anastomosis is the most evidence-based surgical intervention for slow transit constipation, though it carries significant risks of complicatons.[48] In some patients, a 2-stage procedure that begins with a diverting loop ileostomy, which is reversible, may be preferred (Figure 3). This approach allows for informed decisions regarding symptom improvement, with completion colectomy and ileorectal anastomosis potentially offered 6–12 months later for those with significant symptom improvement.[90] Loop ileostomy may also help mitigate the high rates of complication associated with colectomy at the time of stoma creation.[90,91] This approach is particularly valuable for patients who have less definitive indications for colectomy, those who have relative contraindications (including symptomatic regional gut dysmotility), or those at risk for poor functional outcomes.[90] Despite the reported high rates of reoperations and complications following stoma surgeries, many patients opt to retain a temporary stoma because this procedure improved their quality of life.[83,92,93] In selected patients with RC, an ostomy can be a justified last-resort treatment option.

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Funding
Kyle Staller was supported by National Institute of Diabetes and Digestive and Kidney Diseases K23 DK120945

Conflicts of interest
The authors disclose the following: Kyle Staller has served as a consultant for Anji, Ardelyx, GI Health Foundation, GI Supply, Gemelli, Laborie, Mahana, ReStalsis, Salix, Takeda, and Sanofi; and received research support from Ardelyx and ReStalsis. Leila Neshatian has served as a consultant for Mindset Health; has served on a data safety and monitoring board for Cook Myosite; and is a member of the Rome Foundation Anorectal Committee and biofeedback section. Anthony Lembo has served as a consultant for Allurion, Ardelyx, Atmo, Bristol Myers Squibb, GSK, Ironwood, J&J, Salix, Takeda, and Vibrant. Adil Bharucha has received grant support and royalties from Minnesota Medical Technologies; holds a joint patent with Medspira; has equity in Exact Sciences, and Phathom Pharmaceuticals; and has served as a consultant for Becapio.